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SWAPPER (SWAP,75) 10/2/73 JGC/BPC/JBL

/PRESWAPPER, HANDLES RUN TIME FOR RUNNING USER

SWAPT, 0 /TIME OF LAST DEBREAK (IN MS.)

SWAP, DSC 1700 /BREAK ON CHAN 17

SZS 10

JSP CEXECTT

EEM

RCK

LAI

SUB SWAPT

/CALCULATE ELAPSED TIME

SPA

ADD (60000.)

DAC SWAPT

/SAVE ELAPSED TIME

LIF"U"SCF"U"CLL

/PRESERVE LINK IN IO

TAD I (TOTRT+1)

DAC I (TOTRT+1)

SZL"U"LF1

IDX I (TOTRT)

LAC (700000)

AND I ERSTAT

SZA

SAD (400000)

JMP SWAPH

DAP I (PRIORITY)

JMP SWAPI

SWAPH,

LAW 777

/UPDATE UNCHARGED RUN TIME

AND I ERSTAT

DAC RUNTM

DAC SRUNTM

LAW 7777

AND I (PRIORITY)

LIA

ADD SWAPT

DAP I (PRIORITY)

SWP

XAI

RAR 6S

SMA

/USER USED ANOTHER 32 MS YET?

JMP SWAPI

/NO, SO DON'T CHARGE HIM ANYTHING

DZM SWCTR

LAC RUNTM

/CHECK IF USER'S QUEUE IS IN RANGE

LIA

SUB SWAPUL

/LOWEST QUEUE NOT MODIFIED (LUB)

SWP

SUB SWAPLL

/HIGHEST QUEUE NOT MODIFIED (GLB)

SPI"U"SZM I

JMP SWAPL

/NOT IN RANGE, CHARGE 1 TICK

LAC I (PRIORITY)

SAR 6S

SAR 6S

SPQ

SUB (1)

/FORM THE TWO'S COMPLEMENT FOR NEG. PRIORITY

DAC SWCTR

/AND SAVE IN PROPER ISP FORMAT

SPQ

JMP SWAPL

/ZERO OR NEG. PRIORITY, GO CHARGE RIGHT AWAY

	LAW 1740	/SEE IF IT'S TIME TO CHARGE HIM YET
	AND I (PRIORITY)	
	SAR 59	
	SAS SWCTR	
	JMP SWAPI	/NOT TIME TO CHARGE YET
SWAPL,	LAW 37	/TRUNCATE UNCHARGED RUN TIME
	AND I (PRIORITY)	
	DAP I (PRIORITY)	
SWAPM,	LAW I 1	/NOW LOOP CHARGING CLOCK TICKS
	ADD RUNTM	
	LIA	
	AND RUNTM	
	SWP	
	SNI I	
	JMP SWAPA	
	LAC RUNTM	/USER IS CHANGING QUEUE
	SAL 29	
	SUB (1)	
	AND (777)	
	DZM HOTFLG	/SIGNAL THE CHANGE
SWAPA,	DAP RUNTM	/UPDATED RUN TIME
	SAD (777)	/GUY PUT ON BOTTOM OF LOW QUEUE?
	JMP SWAPA1	/YES, LEAVE HIM THERE
	ISP SWCTR	/THROUGH PUNISHING THIS GUY?
	JMP SWAPM	/SOME PEOPLE NEVER LEARN
SWAPA1,	LAW 777	
	LSM	
	AND I ERSTAT	
	SAD SRUNTM	/SKIP IF QUEUE HAS BEEN CHANGED BY...
	LAC RUNTM	/...SOME OTHER SB CHANNEL
	DAP I ERSTAT	
	ESM	
	LAC I NETHND	
	SAS ERSTAT	/IS THIS GUY "NCP" ?
	JMP SWAPI	
	LAC (200000)	
	AND I ERSTAT	
	SZA	
	JMP SWAPI	
	LAW 7777	
	AND I ERSTAT	
	SUB (200)	
	SPA	
	JMP SWAPI	
	LAW 40	/GOOSE "NCP"
	DAP I ERSTAT	

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SWAPI, LIO HOTFLG
SNI
JMP SWAPB
SWDBK, LAC RSTAT
SAS BDSTAT
SZS I 30
JMP SWDBK1

SWAPB, LAC 74
DAC I (AC) /SETUP P POINTERS FROM DEBREAK AREA
LIO 75
DIO I (PC)
JSP C16TSV
LIO 76
DIO I (IO)
LIF"U"SCF"U"CLL
DIO I (FLAGS)
LAW I 7777
AND I ERSTAT
SAD (CLKHNG)
JMP SWORG
CLI
LAC I (BILLTT)
RCR 98
RIL 98
DIO I (BILLTT)
SZA
DAP I ERSTAT

SWORG,	ESM	
	LEM	
	CLC"U"CLF 7	
	DAC HOTFLG	/INIT, HOTFLG
	LAW I 7777	/SEE IF XDDT PERMANENTLY HUNG
	AND STAT 0	
	DAP STAT+0	/XDDT ALWAYS QUEUE ZERO
	LIO BDSTAT	
	SNI I	
	DAP I BDSTAT	/FASTRAND SWAPPER ALWAYS AT QUEUE 0
	SAS (40000)	
	JMP SWL0	
	LAW I 2	
	SAS I DCORE	
	JMP SWL0	
	LAW STAT 0	
	DAC I DCORE	
	DZM DDTGUY	
SWL0,	CLA	
	SAS LDSTAT	
	STF 1	
	SAS BDSTAT	
	STF 2	
SWL0A,	LAW STAT	/INIT TO FIND BEST GUY
	DAP SWV1	
	LAW WHERE	
	DAP SWV2	
	LAC (I)	
	DAC SWBQ	
SWV1,	LIO .	/GET HIS QUEUE
	SPI	/HUNG?
	JMP SWL1	/YES, IGNORE
	LAW 740	/NO
SWV2,	AND .	/FIND OUT WHERE HE IS
	RAR 6S	
	SAD (ADD)	
	LAW 1	
	ADD (SWSTB)	/LOOK UP IN SKIP TABLE
	DAP SWV3	
SWV3,	XCT .	/SKIP IF ACCESSIBLE
	JMP SWL1	/INACCESSIBLE, IGNORE
	CLA	/FIGURE EFFECTIVE QUEUE
	RCL 6S	
	SZA	
	CLI	
	RIR 6S	
	LAI	
	SUB SWBQ	
	SMA	
	JMP SWL1	/NOT AS GOOD AS BEST SO FAR
	DIO SWBQ	/BETTER, UPDATE BEST SO FAR
	LAC SWV2	
	DAP SWV7	
	SNI	/FIND A GUY WITH QUEUE 0?
	JMP SWL37	/YES, SO WE CAN STOP LOOKING
SWL1,	IDX SWV1	/LOOK AT NEXT GUY
	IDX SWV2	
	SUB (AND)	
	SAS WTOP	
	JMP SWV1	

LAC SWBQ
SAD (I)
JMP SWBGC

/DID WE FIND ANYBODY?
/NO, RUN BEST GUY IN CORE

SWL37,	CLC	
	DAC SWWQ	
	LAW CS1	
	DAP SWV4	
SWV4,	LAC .	/GET CORE STATUS
	SPA	
	JMP SWL2	/UNAVAILABLE, IGNORE
	SZA I	
	JMP SWL3	/FREE, STOP SEARCHING
	DAP SWV5	
SWV5,	LIO .	/LOOK AT STATUS OF PGM IN THAT CORE
	SPI	
	JMP SWL9	/HUNG, EFFECTIVE QUEUE IS (I)
	CLA	
	RCL 6S	
	SZA	
	CLI	
	RIR 6S	
SWL10,	LAI	/GET STATUS INTO AC TOO
	SUB SWWQ	
	SPA	
	JMP SWL2	/ALREADY HAVE A WORSE GUY
	DIO SWWQ	/REMEMBER NEW WORSE GUY
	LAC SWV4	
	DAP SWT2	
SWL2,	IDX SWV4	/LOOK AT NEXT CORE
	SAS (LAC CS3+1)	
	JMP SWV4	
	LAC SWWQ	
	SPA	
	JMP SWNN	/DIDN'T FIND ANY CORES AT ALL, WAIT
	SUB SWBQ	
	SPQ	/SWAPPABLE PAIR?
	JMP SWBGC	/NO, GO RUN BEST GUY IN CORE
SWL8,	LAW 740	/YES, WHERE IS NON-CORE GUY?
SWV7,	AND .	
	CLI	
	SZA I	
	JMP SWLD	/LITTLE DRUM
	SAS (40)	
	SAD (100)	
	STF 4	/FG 4 ON MEANS B.D., OFF MEANS LIMBO
SWL35,	SZF 3	/FG 3 INDICATES FREE CORE
	JMP SWFC	
	LAW WHAT	/INIT. TO SEARCH FOR FREE L.D. FIELD
	DAP SWV6	
SWL5,	LAW 777	
SWV6,	AND .	
	SAD (400)	
	JMP SWL4	/FOUND FREE L.D. FIELD
	IDX SWV6	
	SAS (AND WHAT+40)	
	JMP SWL5	
	SZF 4	
	JMP SWL11	/GUY IS ON BIG DRUM
	DAP SWV11	/GUY IS IN LIMBO, INIT. TO FIND B.D. FIELD

```
SWL12,    LAW 777
SWV11,    AND
          SAD (400)
          JMP SWL11A    /FOUND FREE B.D. FIELD
          IDX SWV11
          SAS (AND WHAT+140=16)
          JMP SWL12    /KEEP LOOKING
```

/SORRY, YOU LOSE.

```
SWL36,    LIO I SWV7
          LAW I WHERE-STAT
          ADD SWV7
          DAP SWV11A
          LAW 300
```

SWV11A,

```
LSM
DAC I SWV7
LAC (UNUSED)
```

```
DIP
LAW 200
XAI"U"CLL"U"CML
SAS (700)
JMP SWV11C
EEM
LAC (DCH SWV11Z)
JDA 14STXT
JMP SWORG
```

/WE'LL NEED THE LINK LATER

SWV11C,

```
JDA TTNUM    /SET UP TT POINTERS
DZM I TTPP   /FLUSH THE PROGRAM NUMBER
LAW TTSTPT
DAC SWV11W
```

SWV11D,

```
LCH I SWV11W
SAD (SPO
JMP SWORG
JDA TTS      /GO TYPE OUT THE CHARACTER
NOP
JMP SWV11D
```

SWV11W,

0

SWV11Z,

```
472224      /PSU MISSED
004471
222265
647756
```

/MISCELLANEOUS BITS OF CODING

/FOUND FREE CORE

```

SWL3,      STF 3           /FREE CORE INDICATOR
           LAC SWV4
           DAP SWT2        /REMEMBER WHICH CORE
           JMP SWL8

```

/SKIP TABLE FOR FIGURING ACCESSIBILITY

```

SWSTB,     SZF 1           /ON L.D.
           SZF 2           /ON B.D.
           SZF 6           /IN LIMBO
           NOP             /HALTED PGMS ARE NEVER ACCESSIBLE
           NOP             /CORE IS NEVER ACCESSIBLE
           SZF 6           /STARTUP PROGRAM PRIME

```

/PGM IN CORE IS HUNG

```

SWL9,      LIO (I)
           JMP SWL10
SWV20,     LIO WHAT        /VARIABLE FOR SWL11 LOOP

```

/FIND GUY TO PUT ON B.D.

```

SWL11A,    SZF I 2
           JMP SWL11
           STF 6
           JMP SWL0A

```

```

SWL11,     LAC SWV20
           DAC SWV13
           CLC
           DAC SWWQ

```

```

SWV13,     LIO .           /SEARCH ONLY LD WE HAVE
           LAW 700
           NAI
           SZA

```

```

           JMP SWL14        /NOT A GUY THERE
           LAW STAT
           AAI

```

```

SWL13,     DAP SWV12
           SAD (STAT)
           JMP SWL14

```

```

SWV12,     LAC .           /DON'T SWAP XDDT ON FASTRAND
           LIA             /GET GUY'S STATUS
           AND (720000)    /INTO IO
           SAS (400000)
           SAD (020000)
           JMP SWL14

```

```

           LAW I 7777      /SKIP ANY FASTRAND GUYS
           NAI
           SZA
           CLI
           SPA

```

```

           LIO (I)         /EXCEPT HUNG GUYS
           LAI             /EFFECTIVE Q IN AC,IO
           SUB SWWQ
           SPA

```

```

           JMP SWL14        /WORSER GUY ALREADY (SIC)
           DIO SWWQ        /REMEMBER THIS GUY

```

```

LAW WHERE=STAT
ADD SWV12
DAP SWV16
SWL14, SZF 6 /GET NEXT GUY, CORE OR LD?
JMP SWL15 /CORE, GO LOOK AT NEXT CORE
IDX SWV13 /LOOK AT NEXT LD FIELD
SAD (LIO WHAT+40)
LAC (LIO WHAT)
DAP SWV13
SAS SWV20
JMP SWV13
LAW WHAT
ADD I SWV16
LIO SWWQ
SPI I
DAP SWV20
STF 6 /NO LD FIELDS LEFT, LOOK AT CORES NOW
LAW CS1
DAP SWV15
SWV15, LAC . /CORE STATUS WORD
SZM /IS CORE OK?
JMP SWL13 /YES, COMPARE GUY WITH OTHERS
SWL15, IDX SWV15 /GET NEXT CORE
SAS (LAC CS3+1)
JMP SWV15
LAC SWWQ
SPA
JMP SWBGC /NO, WAIT
SUB SWBQ
SPQ /FASTRAND SWAPPABLE PAIR?
JMP SWBGC
LAW 400 /YES, WHERE IS HE?
SWV16, AND . /WHERE PTR TO GUY FOUND
SZA
JMP SWBD /CORE, GO SET UP BD WRITE
SAS LDSTAT /LD, SETUP LD READ UNLESS LD BUSY
JMP SWBGC /LD BUSY, WAIT
LAW WHAT
ADD I SWV16
DAP SWV13
SWL17, LAW I WHAT
ADD SWV13 /GET L.D. READ FIELD
DAP SWT3
RAR 6S /INTO HI ORDER AC
IOR (4000000) /SET READ BIT
DIP LDIA
LAW 777
AND I SWV13 /GET INCOMING PGM
ADD (STAT)
DAP LDT2 /SETUP FOR L.D. S.B. ROUTINE
ADD (WHERE=STAT)
DAP LDV1

```

```

LAW I CS1-1-400          /GET PHYSICAL CORE
ADD SWT2
DAC LDT1
RAR 4S
DIP LDCL
LAC SWT2
DAP LDV2
SZF 3
JMP SWL19                /READ ONLY, GO DO DIFFERENT THINGS
LAW 201                  /SAVE L.D. READ FIELD FOR NEXT SWAP
DAC I SWV13
LIO SWT3                /GET READ FIELD
LAC SWFLDF              /RESET SAVED L.D. FIELD
DIO SWFLDF
SWL21, LIA                /GET WRITE FIELD IN AC AND IO
RAR 6S
IOR (400000)           /SET WRITE BIT, WRITE ONLY JOINS HERE
DIP LDWC
LAW WHERE=STAT
ADD I SWT2              /GET WHERE PTR. TO GUY BEING WRITTEN
DAP .+1
DIO .                  /SAY HE'S ON THE DRUM
LAW WHAT                /GET WHAT PTR. TO WRITE FIELD
AAI
DAP SWV17
LAW I STAT              /SAY DRUM FIELD HAS GUY
ADD I SWT2
SWV17, SWL23, DAC .
CLC                    /SET UP STATUS FOR SWAPPING
DAC LDSTAT
DAC I SWT2              /SAY CORE IS HELD
LAW I 2
DAC LDERC
LSM
DRA
LAW 48.
AAI
DAP LDIA
DAP LDCL
LIO LDIA
DIA
LIO LDWC
DWC
LIO LDCL
DCL
JMP SWORG

```

/NORMAL ENTRY TO LITTLE DRUM CODING

```

SWLD,    LAW WHAT
          ADD I SWV7      /GET READ FIELD
          DAP SWV13
          JMP SWL17       /ENTER L.D. CODING

```

/READ ONLY CODING

```

SWL19,    LAW 400
          DAC I SWV13     /MARK READ FIELD AS FREE
          CLA             /SETUP TO NOT WRITE
          DIP LDWC
          JMP SWL23       /GO DO DRUMSWAP

```

/WRITE ONLY CODING

```

SWL18,    CLA             /ENTRY POINT, SETUP TO NOT READ
          DIP LDIA
          LAC SWT2        /GET CORE TO SET UP FOR S.B. ROUTINE
          DAP LDV2
          SUB (CS1-1)     /GET PHYSICAL CORE
          RAR 48
          DIP LDCL
          LAC SWV6
          SUB (AND WHAT)  /GET WRITE FIELD
          JMP SWL21       /ENTER MAINLINE SETUP CODING

```

/LITTLE DRUM SEQUENCE BREAK ROUTINE

```

LD,        DRA             /GET DRA
          LAC .            /10 MICROSECS.
          SPI
          JMP LDERR        /ERROR ROUTINE
LDERI,     DZM LDSTAT      /RESTORE STATUS, UNAFFECTED BY EM
          DZM HOTFLG      /MARK ALARM
          ISB 1700        /ACTIVATE SWAPPER
          LAC LDIA
          SMA             /WRITE ONLY?
          JMP LDWO        /YES
          LIO LDT1        /PRE SET-UP CORE OF GUY WHO CAME IN
LDV1,      DIO .          /MARK GUY WHO CAME IN AS BEING IN CORE
LDT2,      LAW .          /STAT PTR. FOR GUY WHO CAME IN
LDV2,      DAC .          /RESTORE CORE STATUS
LDERGI,    LAC 24
          LIO 26
          JMP I 25        /DEBREAK

LDWO,      CLA             /SET CORE TO FREE
          JMP LDV2

LDT1,      0

```

```
LDERR,      IDX LDERRS      /COUNT AN ERROR
            LAC LDIA
            DIP LDERRS      /SAVE READ FIELD
            RIL 18
            SPI I
            XX              /TIMING ERRORS
            ISP LDERC
            JMP LDERG      /TRY AGAIN
            LAW I STAT
            ADD LDT2
            RCR 98
            LAW I 3
            EEM
            LSM
            JDA 14SOCT
            LAC (LDERT)
            JDA 14STXT
            LAC LDIA
            AND (-ADD)
            LIA
            LAW I 2
            JDA 14SOCT
            LAC (LDECRT)
            JDA 14STXT
            ESM
            JMP LDERI
```

```
LDECRT:    775600
LDERT:     436400
           476151
           005600
```

```
LDERC,      0
LDERRS,     0
```

```
LDERG,      CLA
            DIP LDWC
            LAC LDIA
            SMA
            JMP LDWO
            LIO LDIA
            DIA
            LIO LDWC
            DWC
            LIO LDCL
            DCL
            JMP LDERGI
```

/FOUND FREE L.D. FIELD

SWL4, CLA
 SAD LDSTAT
 JMP SWL18

/IS L.D. BUSY?

/NO, DO DRUM WRITE TO GET FREE CORE

/THIS SECTION FINDS THE BEST GUY IN CORE TO RUN

SWBGC, LAW CS1 /INITIALIZE

DAP SWV8

LAC (I)

/WORSE THAN WORST QUEUE

DAC SWBQ

SWV8,

LAC .

/GET CORE STATUS

SPQ

JMP SWL6

/NOBODY THERE

DAP SWV9

SWV9,

LID .

/LOOK AT STATUS OF GUY THERE

SPI

/HUNG?

JMP SWL6

/YES, IGNORE

CLA

/FIGURE EFFECTIVE QUEUE

RCL 6S

SZA

CLI

RIR 6S

LAI

SUB SWBQ

/COMPARE WITH BEST QUEUE SO FAR

SMA

JMP SWL6

/NOT AS GOOD

DIO SWBQ

/BETTER, REMEMBER THIS GUY

LAC SWV8

/REMEMBER CORE

DAP RCORE

LAC SWV9

/REMEMBER GUY

DAP RSTAT

DAP ERSTAT

SWL6,

IDX SWV8

/LOOK AT NEXT CORE

SAS (LAC CS3+1)

JMP SWV8

LAC SWBQ

/NO MORE CORES, DID WE FIND ANYONE?

SAD (I)

JMP SWNN

/NO ONE WANTS TO RUN, WAIT

SAD (777)

JMP SWL26

/YES

SWL30,

LAW I CS1-1

ADD RCORE

/SETUP TOP OF RCORE

RAR 4S

DIP RCORE

SAR 2S

/SETUP TO RENAME

DCH (ADD SWV10)

SWV10,

RNM

LAC RSTAT

SAD BDSTAT

/RUNNING THE FASTRAND SWAPPER?

JMP SWV10B

/YES, SO DON'T CHECK SPP STUFF

LID SPOWN

SAD SPUSER

SNI

JMP SWV10B

/THIS USER NOT WANTED BY SPP LOGIC

```

EEM                                     /SUPER=P-POINTER STUFF
LAC SPCNT
DAC T3
LAC SPCAD
DAC T1
LAC (DCH SPWORD)
DAC T2
LIO SPFLAG
SNI                                     /SPP READ OR WRITE ?
JMP ,+4                               /READ, SO COPY FROM USER TO EXEC CORE
LIO T1                               /WRITE, SO CLOBBER USER'S CORE
DAC T1
DIO T2
LAC I T1
DAC I T2
IDX T1
IDX T2
ISP T3
JMP ,=-5
LEM
LAC SPOWN                             /UNHANG SPP OWNER
SAD BDSTAT
LAW FSTAT
DAP SWV10A
LAC (300000
LSM
SWV10A, AND ,
CLI "U"SWP
SNI I
DIP I SWV10A
ESM
LAW SPDONE
DAP SPX                               /SET SWITCH TO GIVE OWNER R2
DZM SPUSER                           /SO USER WON'T BE FOUND AGAIN
SWV10B, LAW STAT
SAD RSTAT
JMP RUNDOT
EEM
SWV19, JMP SWL32                     /SWITCH
SWL31, LAW SWL32
DAP SWV19                             /RESTORE SWITCH
LAW 200
XOR SWSUTT
LIO C16SU
SAD (700)
LIO C16PSU
DIO I (PC)
DAC I (BILLTT)
DZM I (TOTRT)
DZM I (TOTRT+1)
DZM I (IOPRT)
DZM I (IOPRT+1)
DZM I (BITS)
DZM I (PRIORITY)
LSM
LAC I DAT
LIO I TME
ESM
DAC I (SUPTD)
DIO I (SUPTD+1)

```

SWL32,	LEM		
	LAC I RSTAT	/GET GUY'S STATUS	
	RAL 4S		
	SPA		
	JMP IOPWS1	/THE IOP WANTS HIM	
	RAL 1S		
	SPA		
	JMP DDTWS	/XDDT WANTS HIM	
	RAR 2S		
	SPA		
	JMP BRKS	/HE HIT THE BREAK KEY	
SWL33,	EEM	/HE REALLY WANTS TO RUN	
	LIO I (PC)		
	LAC (170000)		
	NAI		
	SZA		
	SAD (160000)		
	JMP SWL33A		
	HLT	/OPERATOR PUSHES CONTINUE, BUG IS IN EXEC,	
	CLI	/USER IS LOSER - CRASH HIM	
SWL33A,	DIO 7S	/SETUP PC IN DEBREAK AREA	
SWL34,	JSP C16TUS	/UNSAVE CORE 16 IF NECESSARY	
	LIO I (FLAGS)		
	LFI		
	LAC I (AC)		
	DAC 74		
	LIO I (IO)		
	DIO 76		
	JMP SWDBK	/DEBREAK	
SWL26,	LAW STAT	/INIT TO CHANGE ALL 777'S TO 776'S	
	DAP SWV18		
SWL28,	LAW 777		
SWV18,	SAS .		
	JMP SWL29	/NOT 777; IGNORE,	
	LAW 776		
	DAP I SWV18	/STORE 776	
SWL29,	IDX SWV18		
	SUB (SAS STAT-WHERE)		
	SAS WTOP		
	JMP SWL28		
	JMP SWL30		

```

SWFC,      SZF 4           /WHERE IS BEST GUY?
           JMP SWBDR       /READ FROM FASTRAND
           LAW I WHERE-STAT /STARTUP STUFF
           ADD SWV7        /GET STAT PTR.
           DAP I SWT2      /SAY HE'S IN CORE
           DAP RSTAT       /SAY HE'S RUNNING
           DAP ERSTAT
           LAC SWT2
           DAC RCORE       /SAY THIS CORE IS RUNNING
           LIO I SWV7      /GET TT #
           DIO SWSUTT
           SUB (CS1-1-400) /GET 400 CORE PTR
           DAC I SWV7      /REPLACE IN WHERE TABLE
           LAW SWL31       /SETUP SWITCH
           DAP SWV19
           JMP SWL30       /GO RUN GUY

SWSUTT,    0

SWNN,      SZS 10
           JSP CEXECTT
           LAW 374
           AND TTF
           RAR 28
           LIA
           LAC I TTF
           RAR 18
           SPA
           LFI
           CLA
           SAS HOTFLG
           JMP SWNN       /WAIT FOR SOMETHING TO HAPPEN
           JMP SWORG      /SOMETHING HAPPENED, GO HANDLE IT

SWDBK1,    DAC SWLGR      /SAVE UP LAST GUY RUN
           RCK
           DIO SWAPT
           LAC 74
           LIO 76
           ASC 1700
           JMP I 75       /DEBREAK

```

/BREAK KEY HAS BEEN HIT
BRKS,

```

LAC RSTAT      /GUY BEING FASTRAND SWAPPED?
SAD BDSTAT
JMP SWL33      /YES, IGNORE BREAK KEY
LAC (-40000)   /REMOVE BREAK KEY BIT
LSM
AND I RSTAT
DAC I RSTAT
ESM
EEM
LAC I (BITS)   /IS HE RUNNING UNDER IDDT
SMA
JMP BRK1+1     /NO : GO CHECK RESTART MODE
LAW 77         /WAS THE TT THE BILLTT
AND I (BILLTT)
JDA TTNUM      /SET UP POINTERS
LAW .+3        /USE TT ROUTINES TO CHECK
DAP TTCKSX
JMP TTCKS1+2
SAS (2)        /+.3 : IS IT A BREAK
JMP BRK1       /NOT BILLTT : IS USER IN RESTART MODE
LAC I (AC)     /COPY AC,PC,IO,FLAGS
DAC I (HH16AC)
LIO I (IO)
DIO I (HH16IO)
LIO I (FLAGS)
DIO I (HH16FG)
DZM I (FLAGS)  /CLEAR LINK FOR C16 "GETDDT" LOGIC
LAC (-500000)  /REMOVE IDDT AND RESTART BITS
AND I (BITS)
DAC I (BITS)
LAC C16BKK
LIO I (PC)
DIO I (HH16PC)
JMP BRK3

```

BRK1,
BRK1+1,

```

LAC I (BITS)   /IS USER IN RESTART MODE
AND (-100000)
SAD I (BITS)
JMP SWORG      /USER NOT IN RESTART MODE
IOR (40000)    /SET DEBREAK LEGAL BIT
DAC I (BITS)
LIO I (PC)
DIO I (RESTPC)
LAW RESTSU

```

BRK3,

```

DAC 75         /RESET PC TO APPROPRIATE POSITION
LAC RSTAT
SAD SPOWN
DZM SPOWN      /HE'S NOT DOING A SUPER-PEEK ANY MORE
LAW 1          /SET TO HIGH QUE
DAP I ERSTAT
JMP SWL34

```

```

/FASTRAND SWAPPER
SWBD,    SZF I 4          /SWAP OR WRITE ONLY
        JMP SWBDW        /WRITE ONLY
        LAW WHAT
        LIO SWFBDF       /GET SLOT TO WRITE ON (WS)
        AAI
        DAP SWBDV1
        LAC I SWV7       /GET SLOT TO READ FROM (RS)
        DAC SWFBDF       /NEW SWAPPING SLOT
        ADD (WHAT)
        DAP SWBDV2
        LAC I SWV16      /GET 40C WHERE C IS CORE INVOLVED
        DAC I SWV7       /SAY GUY ON RS IS NOW IN CORE
        ADD (CS1-1-400)
        DAP SWV4         /CS PTR TO CORE
        DIO I SWV16      /SAY GUY WHO'S IN CORE IS NOW ON WS
        LAW I STAT
        ADD I SWV4
SWBDV1,  DAC .           /WHAT PTR TO WS
SWBDL1,  LAW STAT        /READ ONLY COMES IN HERE
SWBDV2,  ADD .           /WHAT PTR TO RS
        DAC I SWV4       /SAY CORE HAS GUY WHO USED TO BE ON RS
        LAW 201          /CODE: SAVED FOR SWAPPING
        SZF 3
        LAW 400          /READ ONLY, CODE: FREE
SWBDL2,  DAC I SWBDV2    /SETUP NEW STATUS OF RS
        LAW I WHAT 40    /WRITE ONLY COMES IN HERE
        ADD SWBDV1       /CALCULATE WRITE SLOT
        RCR 9S
        LAW I WHAT 40
        ADD SWBDV2       /CALCULATE READ SLOT
        RCR 9S
        LAW I CS1-1
        ADD SWV4         /GET CORE
        RAR 6S           /POSITION
        DCH (ADD ,+1)    /SETUP TO RENAME THAT CORE TO 0
        RNM
        EEM
        LAC I (AC)
        DAC I (FASTAC)
        LAC I (PC)
        DAC I (FASTPC)
        DIO I (AC)
        LAC C16FSW       /STARTING ADDRESS FOR SWAP
        SZF 3
        LAC C16FIN       /SA FOR READ ONLY
        SZF I 4
        LAC C16FOU       /SA FOR WRITE ONLY
        DAC I (PC)
        LEM
        LAW SWBDL5
        SZF I 4
        LAW SWBDL4       /SPECIAL CODING ON RETURN FROM WRITE ONLY
        DAP SWBDV4

```

```
LAC I SWV4
LSM
DAC BDSTAT
LAC I BDSTAT
DAC FSTAT
ESM
LAW 1
DAC I BDSTAT
JMP SWORG
```

```
/WRITE ONLY ENTRY
SWBDW, LAW CS1-1-400
      ADD I SWV16
      DAP SWV4           /SETUP CS PTR TO CORE
      LAC SWV11          /GET FREE BD FIELD FOUND
      DAP SWBDV1         /SETUP PTR
      JMP SWBDL2
```

```
/READ ONLY ENTRY
SWBDR, LAW WHAT
      ADD I SWV7         /GET RS PTR
      DAP SWBDV2
      LAW I CS1-1-400
      ADD SWV4           /GET CORE PTR
      DAP I SWV7
      JMP SWBDL1
```

```
/ERROR RETURN FROM FASTRAND SWAP
SWFERR, LAW I STAT
      ADD RSTAT
      RCR 9S
      LAW I 3
      LSM
      JDA 14SOCT
      LAC (SWFERT)
      JDA 14STXT
      ESM
```

```
/RETURN FROM FASTRAND SWAPPING (CHANNEL 16)
SWFAST, LAC I (FASTAC)
      DAC 70
      LAC I (FASTPC)
      DAC 71
      LEM
      LSM
      LAC FSTAT
      DAC I RSTAT
      DZM BDSTAT
      ESM
SWBDV4, JMP .           /SWITCH
```

```

SWBDL4,  DZM I RCORE      /WRITE ONLY, MARK CORE AS FREE
        LAW I STAT
        ADD RSTAT
        DAC I SWBDV1      /MARK WS AS CONTAINING GUY
        ADD (WHERE)
        DAP SWBDV5
        LAW I WHAT
        ADD SWBDV1
SWBDV5,  DAP .            /MARK GUY AS BEING ON FASTRAND
SWBDL5,  JMP HANG1        /TELL SWAPPER IN ALL 3 CASES

SWFERT:  662226
        006451
        517756

```

/SUBROUTINES TO SAVE AND UNSAVE CORE 16 VARIABLES IF NECESSARY
/PC IN IO

```

C16TUS,  DAP C16TSX
        LAW SWP"A"7777
        JMP C16TS1

C16TSV,  DAP C16TSX
        CLA
C16TS1,  DAP C16TS2
        RIL 2S
        SPI I
C16TSX,  JMP .
        LAW C16TEM
        LIO (160000
C16TS2,  OPR .            /SWP OR NOP
        DAC C16TSS
        DIO C16TSF
        LAW I 10
        DAC C16TSC
C16TSL,  LAC I C16TSF
        DAC I C16TSS
        IDX C16TSF
        IDX C16TSS
        ISP C16TSC
        JMP C16TSL
        JMP C16TSX

C16TSF,  0                /FETCHING PTR
C16TSS,  0                /STORING PTR
C16TSC,  0                /COUNTER

```

```

IOPWS1,  LAC (-20000)  /CLEAR "IOP WANTS" BIT
          LSM
          AND I RSTAT
          DAC I RSTAT
          ESM
          LAC RSTAT
          DAC ISTAT
          LAW I 1
          DAC I RCORE
          LAC RCORE
          DAC ICORE
          SAR 2S
          IOR (100000)  /RENAME CORE TO 4
          DCH (ADD ,+1)
          RNM
          LAW I 7777
          AND ICORE      /RETURN TO IOP WITH REALCORE IN AC
          EEM
          JMP I IOPWGO    /GO TO CORE 17

CEXECTT,  DAP CEXECX
          LAW TTP"TLAC
          IOR EXECTT
          SAL 2S
          DAP CEXEC1
          ADD (3)
          DAP CEXEC2
          LAW 101
CEXEC1,   DAC .
CEXEC2,   DZM .
          LAT
          AND (77)
          DAC EXECTT
          SAL 2S
          ADD (TTP)
          DAP CEXEC3
          ADD (3)
          DAP CEXEC4
          LAW 105
CEXEC3,   DAC .
          LAW STAT 0
CEXEC4,   DAC .
CEXECX,   JMP .

```

LDIA, 0
 LDWC, 0
 LDCL, 0
 LDSTAT, 0
 BDSTAT, 0
 FSTAT, 0
 SWWQ, 0
 SWBQ, 0
 SWT1, 0
 SWT2, 0
 SWT3, 0
 RUNTM, 0
 SRUNTM, 0
 HOTFLG, 0
 SWCTR, 0
 SWAPUL, 1000
 SWAPLL, 20

/PRESWAPPER COUNTS CLOCK TICKS HERE
 /LOWEST QUEUE PRIORITY SCHEME HAS EFFECT
 /HIGHEST QUEUE PRIORITY SCHEME HAS EFFECT

VARIABLES
 CON,CONSTANTS

FOO, FLEXO FOO PRINT /FOO/

7500/
 DSPTB,

140000 ELGIOT
 140000 DHANG
 140000 TYI
 140000 TYO
 140000 TIS
 140000 TOS
 140000 GTY
 140000 RTY

140000 TTCKS
 140000 TTMODE
 140000 RSMC
 REPEAT 2, 140000 ELGIOT
 140000 PSU
 140000 DELAY
 140000 SPP

REPEAT 10, 140000 ELGIOT

REPEAT 2, 140000 ELGIOT
 140000 DWPP
 140000 FF8
 REPEAT 3, 140000 ELGIOT
 140000 DPEEK

140000 ELGIOT
 140000 DCLR /GRDR GET THE READER
 140000 DRPA /RPA READ PUNCHED TAPE ALPHABETIC
 140000 DRLSRD /RRDR RELEASE THE READER
 140000 DGETPU /GPUN GET PUNCH
 140000 DPPA /PPA PUNCH PUNCHED TAPE ALPHABETIC
 140000 DRLSPU /RPUN RELEASE PUNCH
 140000 ,RCK

REPEAT 30, 140000 ELGIOT

DSPTB+100,

C16TB, REPEAT 100, 360000 10
C16TB+100,

C16BKP, 360000 10
C16BKK, 360000 10
C16HH, 360000 10
C16HLT, 360000 10
C16SU, 360000 10
C16PSU, 360000 10
C16FIN, 360000 10
C16FOU, 360000 10
C16FSW, 360000 10

7714/ /CORE 15 ENTRIES

1SECJA, DCH 1SECA
1SECJB, DCH 1SECB
SORADR, DCH SORX
1MIADR, DCH 1MIX
1SSOCT, 0
1SSOC1, 0
1SSTXT, 0
1SSTX1, 0
DAT, 0
TME, 0
NTINT, 0
PTRBRK, DCH PTRRTN
TELINT, DCH PDP7X
NETHND, 0

/ENTRY TO NET INTERRUPT HANDLER

7740/ /CORE 17 ROUTINES

DATADR, DCH DATX
CONADR, DCH CONX
IOPADR, DCH IOPX
IOPWGO, JDA .
XNUMRRO, JDA .

EXPUNGE TYIHNG, TYOHNG, GTYHNG, EOTHNG, RPAHNG, UNUSED, CORHNG, CLKHNG
EXPUNGE SPPHNG, BITS

START HLT=JMP